

Curriculum Vitae Steven Hoekstra

Contact Details

Address: Buikslotermeerdijk 33
1025 WH, Amsterdam
Tel: +31-6-15111223
Email: s.g.hoekstra@uva.nl

Profile

My research is focussed on macroeconomics viewed through the lens of complex systems. I have gained experience in both equilibrium and agent-based modeling through my academic pursuits. My objective is to use insights of both subfields to find a better understanding of macroeconomic phenomena using theory, large scale computational modeling, and empirical methods.

Interests

Complexity, macroeconomics, bounded rationality, agent-based modeling, networks, distributed computing, heterogeneous agent equilibrium models.

References

Prof. Dr. C.H. Hommes
Faculty of Economics, CeNDEF, University of Amsterdam
Bank of Canada, Tinbergen Institute
E: c.h.hommes@uva.nl

Prof. Dr. F.P. Pijpers
Faculty of Science, Korteweg-de Vries Institute, University of Amsterdam
Statistics Netherlands
E: f.p.pijpers@uva.nl

Education

PhD in Economics - University of Amsterdam 2023 - current
Supervisors: Prof. Dr. C.H. Hommes and Prof. Dr. F.P. Pijpers.
Working paper: A Dutch Agent Based Model with Optimal Consumption

MPhil. Economics and Econometrics - Tinbergen Institute 2021-2023
Specialisation: Complexity, Mathematics, and Computation
Thesis: An Agent Based Model of the Dutch economy
Supervisors: Prof. Dr. C.H. Hommes and Prof. Dr. F. Pijpers.

MSc. Econometrics - University of Amsterdam 2020-2021
Specialisation: Complexity and Economic Behaviour
Thesis: Endogenous Financial Instability via Heterogeneous Beliefs
Supervisors: Prof. Dr. C. Hommes

MSc. Economics - London School of Economics 2019-2020
Specialisation: Public Economics
Thesis: The Inequality of Education using a Two Agent New Keynesian Model
Supervisors: Prof. Dr. F. Cowell

Exchange Programme - Royal Melbourne Institute of Technology 2018-2019
Minor in Finance and Programming

BSc. Economics and Business Economics - University of Groningen 2016-2019
Specialisation: Business Economics
Thesis: The Impact of News Coverage on Security Prices

BSc. Econometrics - University of Groningen 2015-2018
Specialisation: Econometrics
Thesis: Arbitrage Smoothing and Term Structure models

Working papers

A Dutch Agent Based Model with Optimal Consumption

Abstract: We develop an extension of the stock flow consistent agent based model of Poledna et al. (2023) by introducing the household optimization problem of Kaplan et al. (2018) which allows for heterogeneous marginal propensities to consume. We use a neural network approach to recalculate the solution of the optimization problem each time period for all the Monte Carlo simulations totalling to more than 100,000 re-optimizations. The household agents make optimal choices given their perceived laws of motion which we model with an AR(1) rule. The households update their optimal choices each period taking the new information set into account. We analyse the model by means of a forecasting exercise where we compare the model to the base version of the agent based model, a version of the agent based model with a more complex consumption heuristic, and to a benchmark VAR model. We calibrate the model for the Dutch economy and forecast the Dutch economy. The optimal consumption model has better forecast performance relative to the benchmark model, the base and the extended heuristic agent base model.

Work Experience

Dutch Central Bank - PhD Intern April - Dec. 2025

- Implemented Bank of Canada CANVAS agent based model for the Netherlands
- Estimated inflation channels using neural ratio estimation
- Wrote note on wage-price channel using this estimated model.

Central Bureau of Statistics Netherlands - Thesis intern Mar. - Sept. 2023

- Calibrated the agent based model of Poledna et al. (2023 EER) with Dutch register and national accounts data.
- Developed an extension to the agent based model which incorporated optimal consumption.
- Run simulations with both agent based models and analysed internal dynamics of the models.

SEO Economisch Onderzoek - Research assistant 2020-2021

- Conducted research on a project about academic pressure under high school students.
- Conducted interviews with stakeholders in secondary education.
- Set up nation wide survey about academic pressure under high school students.
- Performed statistical analysis on survey outcomes.

Teaching

Graduate

Behavioural macro and finance - Lecturer 2025

Undergraduate

Microeconomics for AE - Tutor 2023 - 2024

Mathematics 2: Linear Algebra - Tutor 2024 - 2025

Mathematics 3: Advanced Linear Algebra - Tutor 2025 - 2026

Achievements

Honoraries

Cum Laude - MPhil. Economics and Econometrics, Tinbergen Institute 2023

Cum Laude - MSc. Econometrics, University of Amsterdam 2021

Scholarships

Full scholarship (merit based) - Tinbergen Institute 2022

Tuition fee waver - Tinbergen Institute 2021 - 2023

Visits and summer schools

Visiting Scholar - IASSA Aug. 2023

Visited Dr. S. Poledna at International Institute of Applied Systems to work on agent based models.

Tools for Macroeconomist: Advanced Tools - University of Oxford Aug. 2021

A doctoral - level course on DSGE models. Topics covered:
Solving and simulating models with heterogeneous agents,
continuous time models, occasionally binding constraints and bounded rationality.

Tools for Macroeconomist: The Essentials - University of Oxford Aug. 2021

A doctoral - level course on DSGE models. Topics covered:
Techniques for numerical analysis, perturbation and projection methods,
Blanchard-Kahn conditions and sun spot solutions theory,
estimation tools like the Kalman filter and Bayesian estimation

Skills

- Programming languages: Matlab, Python, R, Julia, C, SQL, Bash
- High performance computing - Parallel computing (MPI, BSPlib)
- Agent based modeling
- Network methods
- Machine Learning
- Econometrics
- Dynamics Stochastic General Equilibrium modeling
- Dynare (economic modeling package)
- Stata (econometrics package)